

WASP-12b

R-0225

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_241222 · created 2026-07-17T02:32:03+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460666.7643 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1354 +/- 0.0091
Transit depth (Rp/Rs)^2	1.83 +/- 0.25 [%]
Orbital Inclination (inc)	82.48 +/- 2.22
Airmass coefficient 1 (a1)	1.0159 +/- 0.0072
Airmass coefficient 2 (a2)	-0.013 +/- 0.023
Scatter in the residuals of the lightcurve fit is	0.71 %
Best Comparison Star	#3 - [257, 348]
Optimal Aperture	3.95
Optimal Annulus	6.99
Transit Duration (day)	0.0988 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R ^{fit} vs published	0.1354 ± 0.0091 vs 0.1178 (deviation 1.93σ)
Duration fit vs published	0.0988 d vs 0.125 d (deviation 9.03σ)
Mid-time O–C	-1.1 min
Depth SNR	14.9
Residual scatter	0.71 %

Light curve

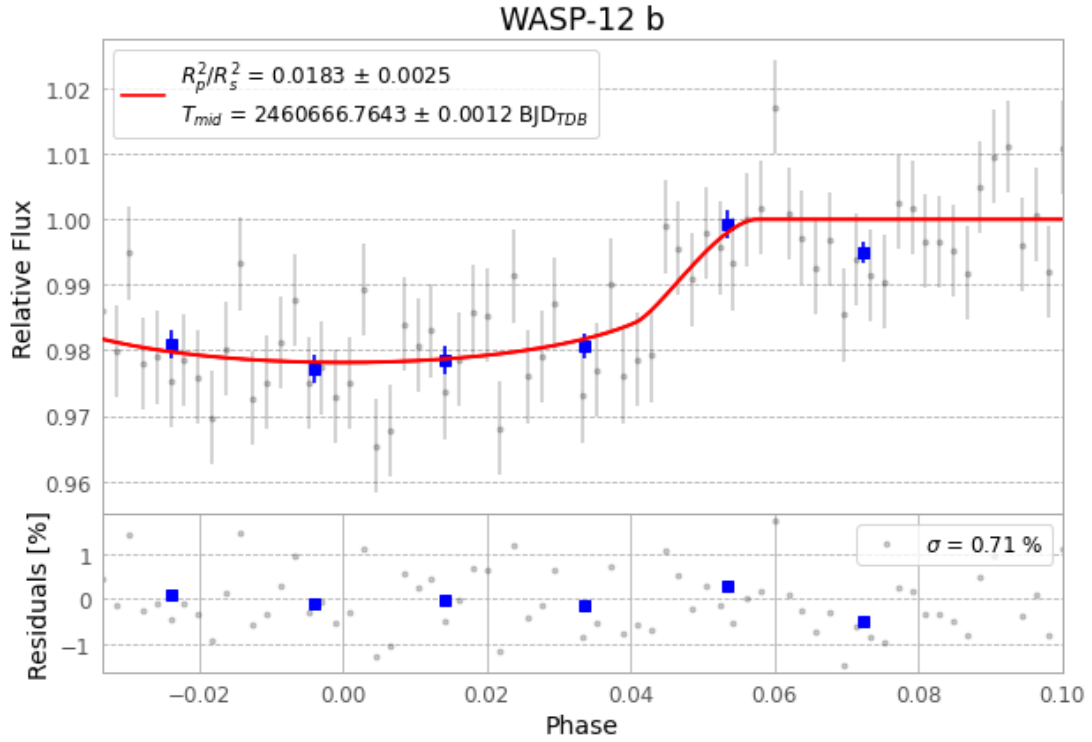


Figure 1 — FinalLightCurve_WASP-12 b_2024-12-21.png (SHA-256 57164c41eb69288b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c8f7b622360c6965...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

71 FITS frames (47 MB), WASP-12241222051816.FITS → WASP-12241222084815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-241222.log (3b27e84cb215...), FinalLightCurve_WASP-12 b_2024-12-21.png (57164c41eb69...), FinalLightCurve_WASP-12 b_2024-12-21.csv (a8f19bdd14b0...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 02:32Z.